

Lost Your Hearing Aid? Just 3D Print It!

With customisation and quick turnaround times becoming crucial in today's production climate, 3D printing is being viewed as the dawn of a new era. Prototype development can not only be speeded up but also monitored, in detail. In fact, 3D printing is expected to shift the focus away from large manufacturing hubs to the designers themselves



Fig. 1: A prosthetic face created using 3D printing (Source: <http://frenchtribune.com>)

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Yes, if you are unable to find your hearing aid, you can simply print it using a 3D printer and, within minutes, you will have a new device. 3D printing has evolved much beyond even recreating something as complex as a hearing aid. In a first-of-its kind procedure, doctors in Britain have created a prosthetic face for a man because the entire left side of his face had to be removed since it was cancerous. Doctors first tried plastic surgery; however, that did not work since the patient was also undergoing chemotherapy and radiotherapy.

The doctors then scanned the left side of the patient's skull and used information technology to develop a new

face. 3D printing was used to create layer upon layer of nylon plastic, with the exact elements that were required in the face reconstruction procedure. Dental surgeons have been using 3D printing since the last few years to create identical models of patients' jaw bones so that they can explore the different surgical processes available to them, prior to a surgery or tooth extraction.

The US President bets on 3D printing

On 12th February, President Barack Obama, in his State of the Union address, talked of ways to revive the US manufacturing economy. He mentioned 3D printing as the possible panacea that could multiply manufac-

turing opportunities. Here is an excerpt from his speech: "Our first priority is making America a magnet for new jobs and manufacturing. After shedding jobs for more than 10 years, our manufacturers have added about 500,000 jobs over the past three years...."

"Last year, we created our first manufacturing innovation institute in Youngstown, Ohio. A once-shuttered warehouse is now a state-of-the-art lab where new workers are mastering 3D printing, which has the potential to revolutionise the way we make almost everything. There's no reason this can't happen in other towns. So tonight, I'm announcing the launch of three more of these manufacturing hubs, where businesses will partner with the Departments of Defence and Energy to turn regions left behind by globalisation into global centres of high-tech jobs. And I ask this Congress to help create a network of 15 of these hubs and guarantee that the next revolution in manufacturing is 'Made in America'."

What is 3D printing all about

3D printing, also known as additive or rapid manufacturing, is a process wherein a three-dimensional solid object of just about any shape can be created from a digital model. The starting point is a digital computer aided design (CAD) file, developed with a 3D modelling program, or scanned into a 3D modelling program with a 3D scanner. The software cuts the design into several hundreds or thousands of horizontal layers to form instructions that the 3D printer understands.